



Mahindra University, Hyderabad
École Centrale School of Engineering
MINOR I Examinations, Fall semester 2025 (2023 Batch)
Program: B. Tech. (Common to CM and NT)
Year: III Semester: I
Subject: Computational Methods for PDE (MA 3115)

Date: 11/09/2025
Time Duration: 1.5 hours

Time: 10:00 -11:30 AM
Max. Marks: 15

Instructions:

1. All the answers that belong to a particular question should be answered in one place in your answer booklet.
2. Scientific calculators are permitted.

Q 1:

Marks: 05

The advection equation $\alpha \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = f(x, t)$, $\alpha \in \mathbb{R}$, is approximated using Lax-Friedrichs scheme,

$$\frac{\alpha(U_{i,j+1} - 0.5(U_{i+1,j} + U_{i-1,j}))}{\Delta t} + \frac{U_{i+1,j} - U_{i-1,j}}{2\Delta x} = f_{i,j}.$$

Then investigate its consistency if (a) $\Delta t = r\Delta x$ and (b) $\Delta t = r(\Delta x)^2$, $r \in \mathbb{R}^+$.

Q 2:

Marks: 05

Reduce the following PDE to the canonical form and find its solution:

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0, \quad 0 < x < 4, \quad t > 0; \quad u(0, t) = 2t, \quad t > 0$$

and

$$u(x, 0) = \begin{cases} x(x-2), & 0 \leq x \leq 2, \\ 2(x-2), & 2 \leq x \leq 4. \end{cases}$$

Q 3:

Marks: 05

Using upwind scheme perform one iteration for Q 2, with $r = 0.5$ and $\Delta x = 0.5$.